

Spatial Guidance in Asynchronous Collaboration: The Impact of Embodied Mixed Reality Instructions on Task Accuracy

Anjela Mayer*
Department of Production
Technology and Systems
Technical University of Sofia

Ines Miguel-Alonso†
Department of Computer
Engineering
University of Burgos

Kevin Kastner‡
CeMOS Research and
Transfer Center
Technical University of
Applied Sciences Mannheim

Andres Bustillo§
Department of Computer
Engineering
University of Burgos

Julian Reichwald¶
CeMOS Research and
Transfer Center
Technical University of
Applied Sciences Mannheim

Jivka Ovtcharova||
Department of Production
Technology and Systems
Technical University of Sofia

Georgi Todorov **
Department of Production
Technology and Systems
Technical University of Sofia

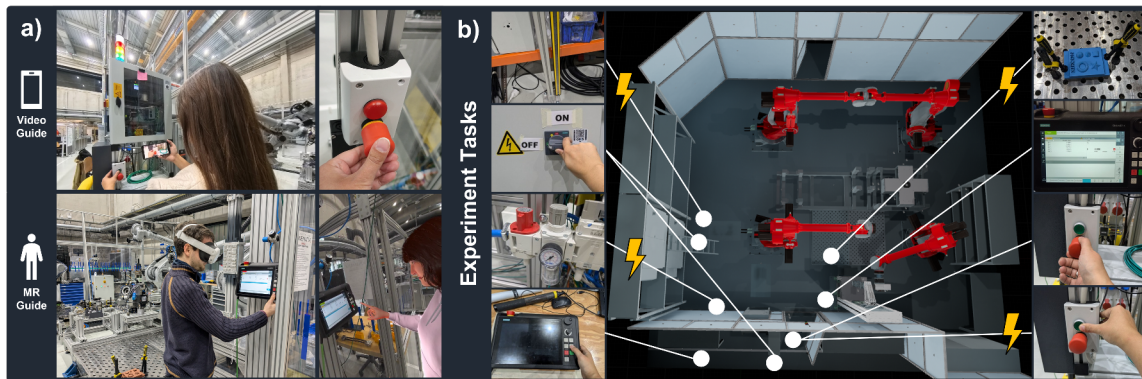


Figure 1: Experimental setup and task overview in a robotic cell commissioning process: a) first-person *Video Guide* (top) and embodied *MR Guide* (bottom); b) Eight commissioning subtasks and their spatial distribution: laser-barrier test, system power-up, pneumatic check, robot unlock (left); workpiece placement, control-panel reset, safety door preparation, safety activation (right). Four subtasks contained seeded omissions/errors to trigger issue reporting and asynchronous expert interaction.

ABSTRACT

Asynchronous collaboration requires instructions that remain interpretable without immediate feedback, yet conventional guidance is often separated from the workspace and may induce errors. A between-subjects study (N=43) compared embodied, spatially registered Mixed Reality guidance (*MR Guide*) with smartphone video (*Video Guide*) in an asynchronous commissioning procedure. Higher task accuracy was observed for *MR Guide* during the initial phase in an unfamiliar workspace, whereas differences diminished later. These findings motivate spatially registered messages and novelty-adaptive guidance strategies in asynchronous collaboration. Future work should quantify attention and examine its relationship with accuracy.

Index Terms: Mixed Reality, Asynchronous Collaboration, Spatial Guidance.

*e-mail: anjela.mayer@tu-sofia.bg

†e-mail: imalonso@ubu.es

‡e-mail: k.kastner@th-mannheim.de

§e-mail: abustillo@ubu.es

¶e-mail: j.reichwald@th-mannheim.de

||e-mail: jivka.ovtcharova@tu-sofia.bg

**e-mail: gdt@tu-sofia.bg

tial Guidance.

1 INTRODUCTION

Asynchronous collaboration is frequently required when collaborators are not available at the same time. In such settings, messages are left for an asynchronous handover and must remain interpretable without immediate feedback or real-time intervention. Ambiguous or context-poor instructions can therefore cause misunderstandings and failures, which is critical in physical work environments (e.g. industrial or medical settings) where safety may be affected.

To document and deliver such messages, conventional Computer Supported Collaborative Work (CSCW) media (e.g. documents or videos) can be used. Such media often detach instructional content from the physical workspace and abstract it into flattened representations. However, physical and procedural workflows rely on spatial references that must be mentally mapped from the medium to the workspace [5]. Therefore, the impact of spatially registered immersive message delivery for asynchronous collaboration is investigated using Mixed Reality (MR).

Prior work has explored the use of MR to support remote and asynchronous instruction by augmenting the workspace with visual guidance. For instance, *AVICol* is an adaptive MR system that provides visual instructions to facilitate remote collaboration [4]. While such approaches demonstrate the potential of MR to improve the interpretability of instructional content, they primarily rely on

adaptive visual overlays rather than embodied instruction.

Several works apply embodied messages in asynchronous MR collaboration [1, 2] and present dynamic and spatially registered embodied collaboration. However, these studies do not assess collaboration quality and do not demonstrate the benefits over non-immersive media.

This work hypothesizes that spatially embedded, embodied MR guidance improves asynchronous collaboration by presenting instructional content in situ, thereby improving clarity and reducing task errors compared to conventional video guidance. Therefore, an embodied MR instruction delivery using in-situ avatar playback (*MR Guide*) is compared to a smartphone-based video playback (*Video Guide*) condition in a realistic industrial guidance scenario.

2 METHOD

A controlled between-subjects study compared two modalities for asynchronous spatial task guidance in a realistic industrial workspace: embodied, spatially registered mixed reality guidance (*MR Guide*, $n = 22$) and smartphone-based first-person video guidance (*Video Guide*, $n = 21$) (Figure 1a). Asynchrony was operationalized via prerecorded expert instructions that participants replayed during execution; to elicit asynchronous interactions, 4 of 8 subtasks contained minor errors or omissions that could be recorded in the corresponding modality and were addressed by expert responses in the subsequent phase.

Conditions. In *MR Guide*, in-situ avatar playback was delivered on a Meta Quest 3 (2064×2208 px per eye). In *Video Guide*, clips were presented in a messenger-style timeline on a handheld Xiaomi Redmi Note 10 Pro (6.67"; 2400×1080) and played on demand.

Procedure. Participants completed the commissioning procedure (Figure 1b) in three parts: Part I: initial execution with prerecorded guidance; Part II: re-execution with access to expert responses to reported issues; Part III: role reversal (recording responses to prerecorded issues). Between parts, participants completed questionnaires in a separate room while the experimenter recorded expert responses for Part II or prepared prerecorded issues reports used in Part III; no real-time interaction occurred.

Measure and analysis. Accuracy was scored per subtask (2=expected completion including required issue reporting; 1=partial or unintended but correct; 0=incorrect/incomplete including missing issue report) and aggregated as a proportion, $\sum_{\text{earned}} / \sum_{\text{possible}} \in [0, 1]$. Between-condition differences were tested using two-tailed Mann–Whitney U tests ($\alpha = .05$).

3 RESULTS AND DISCUSSION

As shown in Table 1, the largest accuracy difference was observed during the initial exposure to the work environment and tasks (Part I), whereas no significant differences were found in the later phases (Part II and Part III). This pattern suggests that embodied, spatially embedded MR instruction may be particularly beneficial when spatial reference frames and procedural mappings are still being established in an unfamiliar workspace. During the experiment, participants were observed to rely solely on the audio while navigating through the robotic cell, rather than pausing or replaying the video when information was missed. These results align with related work in which embodied guidance yielded better learning outcomes than video-based guidance in a virtual reality chemistry lab [3]. Video guidance appears to induce split-attention between the instruction medium and the task environment.

Based on these results, we propose the following design considerations for computer-supported asynchronous collaboration:

1. **Spatial references in messages:** When instructional content includes spatial references, the corresponding information should be spatially anchored within the workspace context. This reduces split attention between the instruction medium

Table 1: Task accuracy by phase (proportion in $[0, 1]$). *MR Guide* shows higher accuracy in Part I and overall; differences diminish in later phases. Two-tailed U tests; $r = |z|/\sqrt{N}$. * $p < .05$, ** $p < .01$.

Measure	MR Guide	Video Guide	U	z	p	r
	M±SD	M±SD				
Part I**	.92±.11	.75±.19	95	3.36	.001	.51
Part II	.90±.17	.94±.12	188	0.62	.530	.10
Part I+II*	.91±.09	.80±.20	128	2.50	.012	.38
Part III	.95±.07	.91±.09	153	1.60	.110	.25
Σ I–III*	.91±.10	.83±.15	138	2.25	.024	.34

and the task environment, thereby helping to prevent errors and learning gaps.

2. **Unfamiliar environments:** For initial exposure to unfamiliar workspaces or tasks, embodied MR guidance should be considered to support spatial grounding and accurate execution of workflows. After users have become familiar with the environment and procedures, simpler forms of instruction may be sufficient.

4 CONCLUSION

In a controlled study of asynchronous spatial task guidance, embodied and spatially registered MR instruction (*MR Guide*) was compared with smartphone video instruction (*Video Guide*). Higher task accuracy was observed for *MR Guide* during initial exposure to an unfamiliar workspace and in the aggregated score, whereas differences diminished after familiarization. These results motivate spatial anchoring for spatially referential instruction content and novelty-adaptive guidance, in which embodied MR guidance is prioritized during onboarding and reduced-complexity representations may be sufficient thereafter. Future work should quantify split-attention directly (e.g. via gaze tracking) and examine its relationship with task accuracy and knowledge gain across modalities.

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